MIL-D-44160(GL)
6 March 1985

MILITARY SPECIFICATION

DETERGENT, LAUNDRY (LIQUID-COLD WATER)

This specification is approved for use by the Natick Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type of a liquid, synthetic, laundry detergent for use in cold water (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-D-43703 - Drums, Shipping, and Storage, Molded Polyethylene

STANDARDS

FEDERAL

FED-STD-536 - Soap and Soap-Products (Including Synthetic Detergents); Sampling and Testing

Beneficial comments, (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7930

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-147 - Palletized Unit Loads

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

Laws and Regulations

16 CPR 1500 - Federal Hazardous Substance Act

(The code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents. US Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 501 - Sampling and Chemical Analysis of Alkaline Detergents

D 3951 - Standard Practice for Commercial Packaging

E 70 - pH of Aqueous Solutions With The Glass Electrode

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of a conflict between the test of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Composition. The detergent shall be formulated to contain the following materials in the indicated amounts. There shall be no kerosine, pine oil, or other organic solvents included in the formulation.

Material	Content, Percent Minimum	Maximum
Nonionic surfactant-linear alcohol ethoxylate	10.0	15.0
Anionic surfactant	2.0	5.0
Complex phosphate water conditioners (calculated as P_2O_5 as per ASTM D 501)	3.0	10.0
Active alkalinity (calculated as Na_2O as per ASTM D 501)	20.0	30.0
Anti-redesposition agents for cotton/synthetic fibers	Shall be present	
Optical brightener (fluorescent dyes)	Shall be present	
Mild perfumes	Use is optional	
Tinting agents	Use is optional	

3.2 Finished product requirements.

3.2.1 Form. The detergent shall be an opaque viscous liquid.

3.2.2 Odor. The detergent shall not have a kerosine-like, rancid, fishy, or acrid odor and shall not impart such an odor to fabrics washed in solutions of the detergent.

3.2.3 Solubility. The detergent shall be completely soluble in water when tested as specified in 4.2.3.

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3.2.4 pH. A 1-percent solution of the detergent shall have a pH of not less than 11.0 and not greater than 11.8 when tested as specified in 4.2.3.

3.2.5 Sudsing. The volume of foam resulting from sudsing shall not be greater than 110 mL when tested as specified in 4.2.3.

3.3 Workmanship. The detergent shall be free of foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspections requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.2 Certification. A certificate of compliance will be acceptable as evidence of conformance with the following requirements:

Requirement	Requirement paragraph
Detergent:	
Composition	3.1
Form	3.2.1
Odor	3.2.2

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4.2.3 Product testing. The detergent shall be tested for the characteristics specified in table I. The lot size shall be expressed in containers. The sample shall be a 1-pint composite made up of detergent selected in equal portions from the number of containers indicated by inspection level S-1. Any test failure shall be cause for rejection of the lot.

TABLE I. Product tests.

Characteristic	Rqmt. para.	Test method	No. of determs.	Results reported as
Solubility	3.2.3	4.3.1	1	Pass or fail
pH	3.2.4	ASTM E 70	2	Avg. to nearest 0.1 unit
Sudsing	3.2.5	4.3.2	1	Nearest 1 mL

4.2.4 Packaging inspection. An examination shall be made to determine that packing and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged. The lot size shall be the number of shipping containers in the inspection lot. The inspection level shall be S-2 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Material	Component missing, damaged, or not as specified.
Content	Not as specified.
Containers	Not as specified.

4.2.5 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

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Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, Location, sequence, or method of application.

4.3 Methods of inspection.

4.3.1 Solubility test. To 1.0 gram of detergent, add sufficient distilled water to make 100 mL of solution. This solution shall be maintained at room temperature (25°C) and stirred vigorously for 5 minutes. The solution shall be inspected immediately for undissolved material. Presence of undissolved material will constitute a failure.

4.3.2 Sudsing test. Determine the sudsing property of the detergent in accordance with Method 2001 of FED-STD-536, with the following modifications.

Dissolve 2.0 ± 0.01 grams of detergent in 1 liter of distilled or deionized water at 60° ± 4°C taking care not to generate a foam. Immerse (for 10 minutes) a 250-mL unstoppered graduated cylinder up to the 150-mL graduation mark in a water bath maintained at 38 ± 2°C. Carefully pour 50 mL of the detergent solution into the preheated graduated cylinder and allow to remain in the bath for 10 minutes prior to shaking. Stopper the graduated cylinder and hold the stopper to prevent the stopper from loosening due to internal pressure build-up, and proceed as directed in Method 2001 of FED-STD-536.

5. PACKAGING

5.1 Packing. Packing shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Detergent shall be furnished in 5, 15 and 55 gallon (as applicable see 6.2), self-supporting molded polyethylene drums, conforming to MIL-C-43703. Ultraviolet light protection is not required for product protection. The five gallon and fifteen gallon drums shall be provided with a reclosable dispenser feature which is of such size and shape to allow the detergent to be dispensed from the container.

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5.1.2 Commercial. Detergent shall be packed in accordance with ASTM D-3951.

5.2 Palletization. When specified (see 6.2), detergent packed as specified in 5.1 shall be palletized in accordance with load type XII or XIII (as applicable) of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the Contracting Officer.

5.3 Marking. In addition to any special marking required by the contract or purchase order, containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D-3951, as applicable.

5.3.1 Labeling. Each unit container of detergent shall be durably and legibly marked on the outside with the following information and precautions, and shall conform to 16 CFR PART 1500. Unless otherwise specified, directions for use shall be marked on the outside of each unit container.

Detergent, laundry (liquid-cold water)

A cold water laundry detergent intended for laundering cottons, cotton/polyester blends, and synthetic fabrics in soft and hard water in Army field and fixed laundries, institutional laundry washwheels.

Caution: Not for use in dishwashing machines, or on aluminum surfaces. Do not use on aircraft surfaces. Not for use in areas where detergents containing phosphates are prohibited.

6. NOTES

6.1 Intended use. The detergent is intended for use in soft, hard, or sea water for cold laundering of cotton and synthetic fabrics.

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6.2 Ordering data. Aquisition documents should specify the following:

a. Title, number and date of this document.
b. Size container required (see 5.1).
c. Selection of applicable level of packing (see 5.1).
d. When palletization is required (see 5.2).

Custodian:

Army - GL

Review activities:

Army - MD
DLA - CT

Preparing activity:

Army - GL

Project No. 7930-A381

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-44160(GL)	Detergent, Laundry (Liquid-Cold Water)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICOP, 1 Apr 84

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